UPES

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QUESTION 1)

CODE:

```
#include <stdio.h>

int main() {
    int n, i, j, edges, u, v;
    int adj[10][10] = {0};

    printf("Enter vertices: ");
    scanf("%d", &n);

    printf("Enter edges: ");
    scanf("%d", &edges);

    printf("Enter edges (u v):\n");
    for(i = 0; i < edges; i++) {
        scanf("%d %d", &u, &v);
        adj[u][v] = 1;
    }

    printf("\nAdjacency Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++)
            printf("%d ", adj[i][j]);
        printf("\n");
    }

    int vertex, in = 0, out = 0;
    printf("\nEnter vertex: ");
    scanf("%d", &vertex);

    for(i = 0; i < n; i++) {
        if(adj[i][vertex]) in++;
        if(adj[vertex][i]) out++;
    }

    printf("In-degree = %d\nOut-degree = %d\n", in, out);

    return 0;
}
```

OUTPUT:

```
Enter vertices: 4
Enter edges: 4
Enter edges (u v):
1 2
2 0
3 2
4 1

Adjacency Matrix:
0 0 0 0
0 0 1 0
1 0 0 0
0 0 1 0

Enter vertex: 2
In-degree = 2
Out-degree = 1



=== Code Execution Successful ===
```

QUESTION2)

```c
#include <stdio.h>

#define MAX 10

int adj[MAX][MAX];
int n;

int isUndirected() {
    int i, j;
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++) {
            if(adj[i][j] != adj[j][i]) {
                return 0;
            }
        }
    }
    return 1;
}

int main() {
    int i, j;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter adjacency matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++) {
            scanf("%d", &adj[i][j]);
        }
    }

    if(isUndirected()) {
        printf("The graph is Undirected.\n");
    } else {
        printf("The graph is Directed.\n");
    }

    return 0;
}
```

CODE:

OUTPUT:

Output

```
Enter number of vertices: 3
Enter adjacency matrix:
0 1 2
1 2 3
0 2 3
The graph is Directed.


=== Code Execution Successful ===
```